

CSE 4615:: Wireless Networks

Future Trends

Lecture 3

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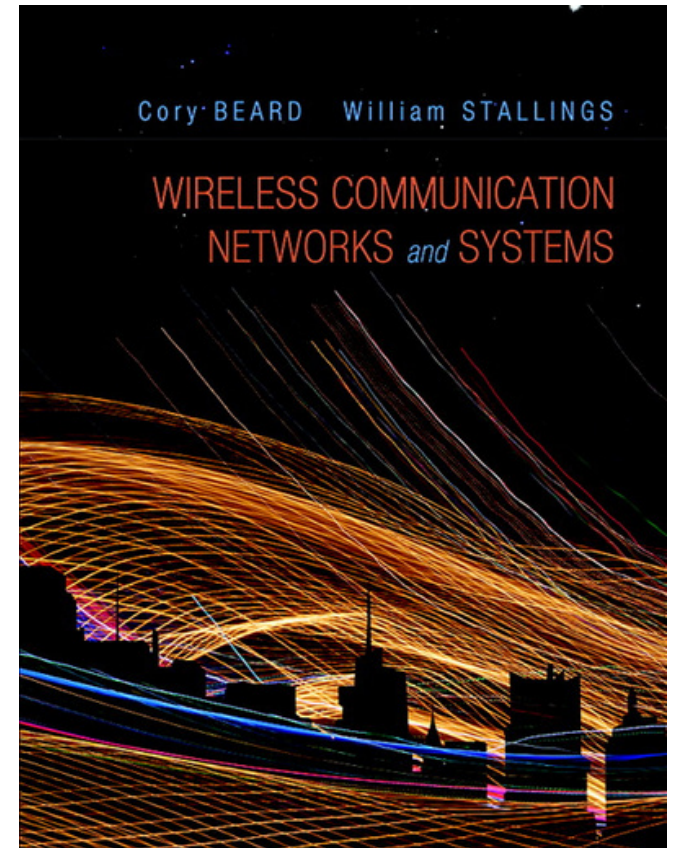
CHAPTER 1

INTRODUCTION

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Wireless Communication Networks and Systems

1st edition

Cory Beard, William Stallings

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Global cellular network

- Generations
 - 1G – Analog
 - 2G – Digital voice
 - Voice services with some moderate rate data services
 - 3G – Packet networks
 - Universal Mobile Phone Service (UMTS)
 - CDMA2000
 - 4G – New wireless approach (OFDM)
 - Higher spectral efficiency
 - 100 Mbps for high mobility users
 - 1 Gbps for low mobility access
 - Long Term Evolution (LTE) and LTE-Advanced

Mobile device revolution

- Originally just mobile phones
- Today's devices
 - Multi-megabit Internet access
 - Mobile apps
 - High megapixel digital cameras
 - Access to multiple types of wireless networks
 - Wi-Fi, Bluetooth, 3G, and 4G
 - Several on-board sensors
- Key to how many people interact with the world around them

Mobile device revolution

- Better use of spectrum
- Decreased costs
- Limited displays and input capabilities
- Tablets provide balance between smartphones and PCs
- Long distance
 - Cellular 3G and 4G
- Local areas
 - Wi-Fi
- Short distance
 - Bluetooth, ZigBee

Future trends

- **LTE-Advanced** and **Gigabit Wi-Fi** now being deployed.

Gigabit Wi-Fi :

- Ultra-fast wireless internet with speeds of **1 Gbps or more**
- Typically supported by **Wi-Fi 5 (802.11ac)** and **Wi-Fi 6 (802.11ax)**

Key Benefits:

- High-speed downloads and streaming
- Low latency for gaming and video calls
- Better performance in crowded networks
- Ideal for
 - **4K video and smart home** use (resolution of about **4000 pixels** across the screen, sharper, clearer, and more detailed pictures — great for large screens),
 - **Online gaming**

Future trends

- Machine-to-machine communications
 - The “Internet of Things”
 - Devices interact with each other
 - Healthcare, disaster recovery, energy savings, security and surveillance, environmental awareness, education, manufacturing, and many others
 - Information dissemination
 - Data mining and decision support for Big Data
 - Automated adaptation and control
 - Home sensors collaborate with home appliances, HVAC systems, lighting systems, electric vehicle charging stations, and utility companies.
 - Eventually could interact in their own forms of social networking

Future trends

- Machine-to-machine communications
 - 100-fold increase in the number of devices
 - Type of communication would involve many short messages
 - Control applications will have real-time delay requirements
 - Much more stringent than for human interaction

Future trends

- Future networks
 - 1000-fold increase in data traffic by 2020
 - 5G – Started globally around **2019**
 - **High Speed:** Up to 10 Gbps
 - **Low Latency:** ~1 ms
 - **Massive IoT Connectivity**
 - **High Capacity in Crowded Areas**
 - **Network Slicing for Custom Services**
 - **Energy Efficient and Reliable**
 - **Ultra-low latency and massive IoT connectivity**
 - Foundation for smart systems, IoT, AR/VR, autonomous vehicles, and Industry 4.0.

Future trends:

Future networks

6G – Still in the research and development phase

Key Points:

- **Target Launch:** Around **2030** (globally expected).
- **Goal:** 10–100× faster than 5G (up to **1 Tbps** theoretical speeds).
- **Features:**
 - Ultra-low latency (~0.1 ms)
 - AI-integrated Semantic Communication
 - Real-time holograms & XR (Extended Reality)
 - Satellite-terrestrial integration
- **Research Countries:** USA, China, South Korea, Japan, EU nations are actively investing in 6G R&D.

Future trends

AI-integrated Semantic Communication



Future trends

- Future Technologies
 - Network densification – many small cells
 - Device-centric architectures - focus on what a device needs
 - **Millimeter wave (mmWave)** - frequencies in the 30 GHz to 300 GHz bands
 - Have much available bandwidth.
 - But require more transmit power and have higher attenuation due to obstructions
 - **OFDM (Orthogonal Frequency Division Multiplexing)**
 - **MIMO (Multiple-Input Multiple-Output)**
 - Native support for machine to machine communication
 - Sustained low data rates, massive number of devices, and very low delays.

Political difficulties

- Between companies

- Need common standards so products interoperate
- Some areas have well agreed-upon standards
 - Wi-Fi, LTE
 - Not true for Internet of Things technologies

- Spectrum regulations

- Governments dictate how spectrum is used
 - Many different types of uses and users
- Some frequencies have somewhat restrictive bandwidths and power levels
 - Others have much more bandwidth available